

Survival Node

USER GUIDE

Eastern Shore AI, LLC

support@easternshore.ai

www.easternshore.ai

PO BOX 317 · Parsonsburg, MD 21849

INITIAL SETUP

Do the "First Steps" below immediately after delivery.

Your Survival Node is built around an **offline-first philosophy** — the maps, AI models, and reference libraries on this device are designed to work without internet access, cell signal, or a service account. You do not need a SIM card or service plan for any of its core capabilities.

About Your Packaging

Your Survival Node ships in a standard cardboard box which may display the logo or branding of another company. This is standard supply chain practice and does not indicate any partnership or affiliation between that company and Eastern Shore AI, LLC.

What you'll find inside the hard case: The phone arrives wrapped in protective bubble wrap packaging. The solar battery arrives in its original manufacturer packaging (opened, inspected, tested, and repacked before shipment). The two Mission Darkness Faraday bags will be in their original packaging and placed under the hard case inside the shipping box — not inside the case itself. Complete the First Steps below, then place the phone and solar battery into their Faraday bags before sealing the case for storage.

What's in the Box

- Survival Node phone — debloated Android OS, optimized for performance and longevity, pre-loaded with 50GB+ survival apps and 2 offline LLMs — wall charger, USB-C cable, and shockproof phone case included
- Premium 24,000 mAh Solar Battery — rugged, drop- and splash-resistant — with 3 attached cables (2× USB-C, 1× iOS) plus a USB-A output on the side, built-in flashlight, and emergency strobe
- (2) Mission Darkness Faraday Bags — premium EMP-rated signal-blocking, one for phone, one for solar battery
- Weatherproof hard storage case with padlock and silica gel packs
- Full Survival Node User Guide — available as an app on the home screen (it checks for guide updates whenever you open it on Wi-Fi); the standard PDF version is also pre-loaded on the device in the Download folder

Your home screen: Every preloaded app is on one screen, in A to Z order. Settings is in the bottom row between Phone and Camera, and the phone's built-in tools, such as Clock, Calculator, Recorder, FM Radio and Contacts, are in the Tools folder.

First Steps

Note: Your Survival Node phone was intentionally drained to approximately 30% before shipping to comply with lithium battery shipping regulations. This is normal — don't be surprised if the phone is at 30% or below when you open the box.

- **Charge the solar battery:** Plug your solar battery into wall power using the included charger. You can charge to 100% (all 4 blue LEDs solid) to verify it's working correctly, then bring it back down to **2–3 of the 4 blue LEDs (50–75%)** before sealing for long-term storage.
- **Charge the phone:** Plug the phone in using the included wall charger and USB-C cable. Charge it to about 60%, then power it completely off. Do not store it fully charged or fully drained.
- **Seal for readiness:** Place the phone and power hub into their Faraday bags, fold the top twice, seal tightly, then store both in the case with the included reusable silica gel packs.

Storage Conditions



Store your Survival Node in a cool, dry location — ideally between 60°F (15°C) and 85°F (30°C). Avoid cars, attics, and garages where temperatures reach damaging extremes. The included silica gel packs absorb moisture and are reusable; recharge them when the **blue indicator dot turns light pink**. To

recharge, place the pack with the indicator dot facing up and microwave it for **3–5 minutes**. Once the dot turns blue again, the pack is ready to reuse. Store the case upright, away from direct sunlight.

Silica Pack Safety: Follow the manufacturer guidance when recharging. Keep the indicator side facing up, monitor the pack while heating, and do not exceed the manufacturer's stated heat limit of **210°F**. If a pack appears damaged, leaking, or overheated, discard and replace it.

SIM Card & Cellular Connectivity

No SIM card is included with your Survival Node, and no activation code, carrier plan, or cellular service of any kind is provided. This device has been deep-debloated and OS-modified for offline, off-grid use — it is **not designed to be used with a cellular carrier**. We have not tested it with any carrier or SIM card, and calls, texts, and emergency calls (including 911) may not work even with a SIM installed. Do not purchase expecting cellular voice or data service. Your Survival Node is built for its offline capabilities only.

All core features — AI chat, offline maps, and reference libraries — work fully without any SIM card, carrier connection, or internet access. Don't count on it to call or text for help in an emergency — keep a regular phone for that.

MAINTENANCE (6-MONTH RULE)

Twice per year, run a full readiness check:

- Open the case and remove items from Faraday bags.
- Charge the power hub to **2–3 of the 4 blue LEDs (50–75%)** on wall power before resealing. You can charge to 100% (all 4 LEDs solid) to verify full capacity, then bring it back down to 2–3 LEDs for storage.
- Power on the phone for 10 minutes, charge it back to about 60% if it's lower, then power it off.
- Inspect the battery area — if the phone back feels warped or the battery appears swollen, do not store it. See the Power & Solar Charging section for safe next steps.
- Inspect cables, seals, and Faraday bag folds for wear or damage, then reseal.
- Check the silica gel pack — recharge it if the blue indicator dot has turned pink.
- Open the User Guide app while on Wi-Fi. If an update note appears at the top, tap **Update** to install the newer guide.
- Set the next reminder now — preparedness fails when maintenance slips.

Update Your PocketPal AI Models

AI models improve regularly. Your 6-month check is a good time to see if newer, more capable models are available. Updating takes about 5–10 minutes on Wi-Fi and keeps your offline AI current.

Your Survival Node ships pre-loaded with two offline models: Qwen3.5-4B as the current primary and Qwen2.5-3B as the faster backup. You can add additional models at any time through PocketPal — just be mindful of available storage. When updating, you can swap in newer, more capable models as they become available.

1. Connect the phone to Wi-Fi.
2. Open the **PocketPal** app.
3. Tap the menu (≡) at the top left, then **Models**.
4. Browse available models. Look for updated versions of models you already have, or newer options at similar or smaller sizes (e.g., a newer Qwen or Gemma release).
5. Tap **Download** on any model you want to add.
6. Once downloaded, tap **Load** on the model to confirm it loads correctly.
7. If you are satisfied with the new model, you can delete the older version: tap the menu (≡) → **Models**, then tap the **Delete** icon on the old model to free up storage.

Tip: When replacing a model, swap like-for-like by size — replace your old small model with a new small model, your old medium with a new medium, etc. This keeps your storage balanced and ensures you always have the right tool for every situation.

Update Your Offline Maps

Map data — roads, points of interest, routing — is updated periodically by app providers. Updating your offline maps during your 6-month check ensures your navigation stays accurate.

1. Connect the phone to Wi-Fi.
2. Open **Organic Maps**.
3. Go to **Menu → Maps & Data** or **Settings → Downloads** (exact label varies by app).
4. Look for a **"Check for map updates"** or **"Download updated maps"** option.
5. Select the regions you use and tap **Update**. Downloads can be large — use Wi-Fi only, not mobile data.
6. Once updated, run a quick test search or route to confirm the new data loaded correctly.
7. Disconnect from Wi-Fi when finished.

Update Your Other Apps

While you have Wi-Fi connected, take a few minutes to update your other offline apps as well. Bug fixes and new features make a real difference in the field.

1. Open **F Droid**.
2. Tap the **Updates** tab at the bottom of the screen.
3. Tap **Update All** to apply all available updates at once, or tap individual apps to review and update one at a time.
4. Once all updates are complete, disconnect from Wi-Fi and return to offline mode.

Update the User Guide App

The User Guide app keeps the whole guide on your phone, so it works fully offline. Whenever you open it while connected to Wi-Fi, it checks for a newer version of the guide. Offline, it simply opens as usual.

1. Connect the phone to Wi-Fi and open the **User Guide** app.
2. If a newer version exists, an orange **Guide update available** note appears at the top. Tap **Update**.
3. When the download finishes, tap the file in the notification bar or in **File Manager → Download**.
4. If Android asks, turn on **Allow from this source**, go back, and tap **Update**.
5. Open the guide again. The version number is at the bottom of the page.

The update installs over your current guide; nothing else on your phone changes. If no note appears, you already have the latest guide.

Note: Only connect to a trusted private Wi-Fi network (your home network) during maintenance. Avoid public Wi-Fi when updating your Survival Node.

EMERGENCY DEPLOYMENT

Grid-down protocol:

- After removing the phone from Faraday storage, keep **Airplane Mode ON** to prevent battery drain from constant tower searching.
- Use your offline stack first: maps, local AI, and local knowledge library.
- Conserve power by lowering brightness and closing background apps.

Use-case order: navigation → medical/repair knowledge → communications when needed. After each use, top the phone back up to about 60% from the solar battery, then fully power it down to preserve the battery.

Calls & Texts

The Survival Node isn't designed for calls or texts and hasn't been tested with a SIM card or carrier, so don't plan on using it to call for help — keep a regular phone for that. Keep in mind:

- Your Survival Node's AI, maps, and reference libraries do not require a signal — they run entirely on the device regardless of connectivity.
- For off-grid messaging, the preloaded Briar app can message other Briar users nearby over Bluetooth or Wi-Fi, with no cell service or internet.

POWER & SOLAR CHARGING



Solar Charging — What to Expect: The integrated solar panel is not designed to keep the phone charged during all-day normal use — and that's fine, because this is not a normal-use device. Under emergency conditions — where you use the phone for a specific task and then power it off — the solar battery can sustain the kit for an extended period of time. Eastern Shore AI founder Chris Gorham has powered his test Survival Node exclusively via solar for over a year without ever connecting it to wall power.

The rule: use it, then shut it off. Solar input accumulates while the phone is powered off. Brief emergency sessions draw far less power than the panel can replenish over a day of sunlight. The more disciplined you are about powering the phone off between uses, the longer your solar reserve lasts.

Daily solar routine:

1. During daylight, leave the solar battery panel-up in direct sun to charge.
2. Use the phone for **30 minutes at most**.
3. Charge the phone back to **about 60%** from the solar battery.
4. Power the phone off and seal it back in its Faraday bag.

- Place panel in direct overhead sun and re-angle as sunlight shifts.
- Power the phone **off** when not actively in use — this is the single most important habit for solar-only operation.
- Use wall charging whenever available to top off the reserve.
- Avoid high-heat charging sessions with phone connected.
- Plan your power budget: prioritize mission-critical tasks, then shut down.

Battery Temperature Limits

Lithium batteries are sensitive to heat and cold. Operating or charging outside safe ranges causes permanent capacity loss and, in severe cases, safety risks.

- **Do not charge above 113°F (45°C).** If the power hub or phone feels hot to the touch, stop charging and move to shade before resuming.
- **Do not store below 14°F (–10°C).** Extended freezing storage degrades cells. If your kit has been in the cold, allow it to warm to room temperature before charging.
- Ideal long-term storage temperature: 60°F–75°F (15°C–24°C) at about 60% charge.

Swollen Battery Warning

If you notice the phone back warping outward, or the power hub casing is deformed, stop using the device immediately. A swollen lithium battery is a fire and puncture hazard. Do not attempt to charge it, compress it, or dispose of it in regular trash. Take it to a certified battery recycling location (Best Buy, Home Depot, and many auto parts stores accept lithium batteries at no charge).

Manufacturer Safety Notices

- Do not charge and discharge simultaneously — do not connect input and output ports at the same time, as this will reduce product service life and may prevent connected devices from charging.

- Keep this product dry. Rain, moisture, and liquids will corrode electronic components and circuitry.
- Do not attempt to disassemble this product.

BATTERY MANAGEMENT

Proper battery habits extend device life significantly — especially for a kit that may sit in storage for months before it's needed.

Daily Use Tips

- **Avoid charging to 100% regularly.** Keeping your phone between 20–85% puts less stress on the battery cells and extends overall lifespan. Use Battery Bot Pro to set a charge alarm at 85% so you know when to unplug.
- **Avoid deep discharges.** Repeatedly draining to 0% accelerates degradation. Recharge before you hit 15%.
- **Charge at room temperature.** Charging a hot or cold phone causes permanent capacity loss. Let the device reach room temperature first.
- **Use the original or quality USB-C cable.** Cheap cables can undercharge or intermittently disconnect, which stresses the battery controller.

Long-Term Storage

If storing your Survival Node for an extended period (months to a year+), follow these steps before sealing it in the hard case:

1. Charge the phone to **about 60%**. Do not store it fully charged or near empty.
2. Power the phone completely off before storing.
3. Charge the solar power hub to **2–3 of the 4 blue LEDs (50–75%)** before storage.
4. Store in a **cool, dry location** between 60°F–75°F (15°C–24°C). Avoid attics, vehicles, or anywhere with temperature extremes.
5. **Check every 3–6 months.** Power the phone on briefly to verify charge level hasn't dropped significantly. Top it back up to about 60% if needed, then power back off.

A battery stored at 100% or 0% for months will lose permanent capacity. Storing the phone at about 60% is the single most impactful habit for kit longevity.

AccuBattery — Pre-Installed App

AccuBattery is installed on every Survival Node. Eastern Shore AI uses it for the one battery health test each unit gets before shipping, and the result is linked in your shipping email. AccuBattery remains on your device and is available if you ever want to view battery stats or monitor charge behavior, but no ongoing testing is required from you.

Battery Bot Pro — Pre-Installed App

Battery Bot Pro is a lightweight, open-source battery monitor pre-installed on every Survival Node. It runs in the status bar and notification shade, giving you real-time battery data without opening any app.

- **Status bar percentage** — shows exact charge % in the status bar at all times
- **Live metrics in notifications** — temperature, voltage, health status, and time since plugged/unplugged
- **Charge current (mA)** — shows how many milliamps are flowing to or from the battery (device-dependent)
- **Custom alarms** — set alerts for specific charge thresholds (e.g., alarm at 85% to stop overcharging, alarm at 15% before deep discharge)
- **Temperature alarms** — get notified if the battery temperature exceeds a set limit during charging
- **Battery log + CSV export** — logs charge, voltage, temperature, and health over time; exportable for review
- **Home screen widget** — large 4×1 widget showing live stats at a glance

Recommended Battery Bot Pro settings for Survival Node:

- Set a **charge full alarm at 85%** to prevent overcharging during daily use
- Set a **low battery alarm at 15%** to avoid deep discharge
- Set a **temperature alarm at 105°F (40°C)** to catch overheating before it causes damage
- Enable **battery logging** so you have a health record for the device over time

FENNEC BROWSER

What it is: Fennec is Mozilla Firefox for Android with telemetry and proprietary code removed. It is your Survival Node's default browser for accessing locally cached content, saved HTML files, or any web content when brief Wi-Fi access is available.

Key features:

- Built-in tracking protection — blocks 2,000+ known trackers by default
- No Mozilla telemetry or background data collection
- Add-on support: uBlock Origin and other privacy extensions work natively
- Bookmark manager for offline resources you want one-tap access to

Basic usage:

1. Open **Fennec** from the home screen
2. Tap the address bar and type a URL or search term
3. Access bookmarks via the bottom menu → Bookmarks tab
4. Tap the three-dot menu (⋮) → Settings to adjust tracking protection and default search engine

Tip: bookmark your most-used offline resources so they're one tap away during an emergency.

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ORGANIC MAPS

What it is: Organic Maps provides full offline navigation for the entire United States using OpenStreetMap data. The complete US map pack is pre-downloaded on your Survival Node — no cell signal, no internet, and no Google required.

Key features:

- Full United States map pre-loaded — ready to use with no downloads
- Turn-by-turn voice navigation — driving, cycling, and pedestrian modes
- Offline address search, POI lookup, and route planning
- GPX track import/export for custom routes
- Zero tracking — fully private, no data sent anywhere

Basic usage:

1. Open **Organic Maps** — the full US map loads with no internet connection
2. Tap the search icon (🔍) to find a destination by address or place name
3. Tap **Directions** to plan a route — choose driving, cycling, or walking mode
4. Tap **Go** to start turn-by-turn navigation using your device GPS
5. To drop a pin manually: press and hold anywhere on the map

Tip: GPS lock may take 30–60 seconds in a new location — power the app on before you need to move. Test your destination search and routing before an emergency, not during one.

See the **Maintenance** section for how to update your map data when Wi-Fi is available.

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POCKETPAL

Included in: Every Survival Node (2 offline LLMs)

What it is: PocketPal is an offline AI chat app that runs Large Language Models directly on your device — no internet, no cloud, and no data leaving your phone. It is your Survival Node's core AI tool for survival, medical, and engineering questions.

Model selection as of September 12, 2026:

- **Primary model and the one that powers the Survival Node AI Agent: Qwen3.5-4B-Q4_K_M** — selected for deeper, more capable emergency answers on the 8GB phone. It is slower while generating, so a long response may take several minutes.
- **Fast backup: Qwen2.5-3B-Instruct (Q5_K_M)** — the previous primary and the faster option when time or battery matters. Keep it available for quick answers and as a second model to compare against.

Your Pre-Configured Survival Node AI Agent

Every Survival Node ships with a custom Pal named **"Survival Node"** already configured in PocketPal. It is tuned specifically for survival, field engineering, and emergency response — direct answers first, critical safety warnings in bold, and every response formatted in numbered steps and bullet points. No setup required.

How to use the Survival Node Pal:

1. Open **PocketPal**
2. Tap the **upward-pointing arrow (↑)** to the left of the "Type Your Message Here" input — this opens the Pal picker
3. Tap the **Pals** tab
4. Select **"Survival Node"**
5. Start chatting — the system prompt is applied automatically
6. To ask out loud, switch to Qwen 2.5 first (tap ↑, then the Models tab), then tap the mic next to the send button. It works offline. Voice doesn't work with Qwen 3.5, which needs too much memory

Benchmark Performance (PocketPal, tested on-device)

- **Qwen3.5-4B-Q4_K_M (9/12/2026):** 1.41 t/s token generation · 10.78 t/s prompt processing · Peak memory: 5 GB / 8 GB (59.6%) · 2.73 GB model / 4.21B parameters — current primary, selected for answer depth
- **Qwen2.5-3B (Q5_K_M, 3/26/2026):** 3.50 t/s token generation · 16.10 t/s prompt processing · Peak memory: 5 GB / 8 GB (58.9%) — faster backup

Tip: token generation speed is what you feel while the model is typing its response. Prompt processing speed affects how fast it starts. Qwen3.5 is the current primary because this kit prioritizes answer depth over typing speed; use Qwen2.5 when you need a faster response. Verify critical information against WikiMed or the Survival Manual before acting.

Tip: smaller model = faster and more battery-efficient. Larger model = deeper answers. Switch models from the same Pal picker.

The full system prompt is saved to the **Documents** folder on the device, and a copy is also pinned to the home screen alongside the other pre-loaded apps for quick access. It reads:

```
You are a high-level technical survival and field engineering advisor. Your goal is to provide objective, concise, a

Operational Rules:
1. No Fluff: Do not use introductory phrases. Start every response with the direct answer.
2. Prioritize Safety: List a CRITICAL SAFETY WARNING in bold for dangerous tasks.
3. Format for Clarity: Use numbered steps and bullet points.
4. Density: Provide specific technical measurements.
5. Offline Context: Assume zero internet and limited tools.
6. No Speculation: If factual basis is missing, state DATA NOT AVAILABLE.
```

AI Accuracy & Limitations

AI models are powerful but not infallible. They can confidently state wrong information — this is called "hallucination." Always cross-reference AI outputs against your offline reference libraries before acting on them, especially for medical, legal, or safety-critical decisions. AI is a research tool — not a substitute for professional expertise.

If the AI gives a clearly wrong answer: start a fresh chat session, rephrase your question, or try a larger model.

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F-DROID APP MARKETPLACE

What it is: F-Droid is an open-source Android app marketplace for free, privacy-respecting software. It is your primary source for installing and updating apps that don't require Google Play. All apps listed in the F-Droid catalog are open-source and audited for trackers before listing.

Key features:

- No Google account required — completely anonymous
- All apps are open-source and tracker-audited before listing
- Automatic update notifications for installed F-Droid apps
- Works fully on debloated Android OS without Google Play Services

Basic usage:

1. Open **F-Droid** from the home screen
2. Tap **Latest** to browse recently updated apps, or use the search bar
3. Tap an app → **Install** to download and install it
4. Tap the **Updates** tab (bottom) to see and apply updates to your installed F-Droid apps

Note: F-Droid requires Wi-Fi to download or update apps. All apps on your Survival Node are pre-installed — F-Droid is there for future installs and updates when you have connectivity.

OS & App Compatibility

Every Survival Node ships with a debloated Android OS, optimized for performance and longevity. Some apps requiring Google Play Services — banking apps, certain navigation apps, hardware attestation — may not function. This does not affect your offline preparedness stack. Do not install or turn on Google Play Services, as it would undermine the privacy architecture of your Survival Node. Motorola may install Android updates when the phone is on Wi-Fi. Only connect to Wi-Fi you trust.

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SURVIVAL MANUAL APP

What it is: The Survival Manual app is a fully offline reference guide based on the US Army Field Manual FM 3-05.70 — covering wilderness and emergency survival from shelter and fire to first aid and navigation. Open-source, no internet required.

What it covers:

- **Psychology of Survival** — maintaining mental resilience under stress
- **Shelters** — building emergency shelter from natural and found materials
- **Water** — finding, collecting, and purifying water in the field
- **Fire** — fire-starting techniques with limited or improvised tools
- **Food** — identifying edible plants, trapping, and field food preparation
- **First Aid** — emergency field medical procedures
- **Signals** — visual, audio, and electronic signaling for rescue
- **Navigation** — land navigation without GPS using terrain and stars
- **Dangerous Plants & Animals** — threat identification by region

Basic usage:

1. Open **Survival Manual** from the home screen
2. Browse chapters using the main menu — all content loads offline immediately
3. Use the search function to find specific topics fast

Tip: browse the app before you need it so you know where key chapters live. Fire-starting, water purification, and first aid are the sections most commonly needed first.

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KIWIX

Included in: Every Survival Node (full library)

What it is: Kiwix is an offline content reader that lets you browse large reference libraries — encyclopedias, health guides, instructional content, and more — stored as compressed ZIM files on your device. No internet required.

Basic usage:

1. Open **Kiwix** from the home screen
2. Your pre-loaded libraries appear in the **Library** tab — tap any to open it
3. Browse and read exactly like a website — all offline
4. Use the search bar within each library to find specific topics
5. Tap the back arrow to return to the library list

If a library doesn't appear: swipe down in the Library tab to trigger a storage scan. ZIM files must be in internal storage to be detected.

Pre-Loaded Kiwix Libraries

WikiMed Medical Encyclopedia (Full)

The complete Wikipedia Medical encyclopedia snapshot — thousands of articles covering human anatomy, diseases, symptoms, medications, procedures, and emergency care. Fully searchable. Cross-reference symptoms, understand diagnoses, or look up medication interactions when professional care is unavailable.

Additional Medical Libraries

- **MDWiki Medical Encyclopedia** — healthcare articles curated by WikiProjectMed
- **MedlinePlus** — trusted health information on conditions, tests, and medications
- **Project Gutenberg Medicine** — classic public domain medical books
- **Medicine LibreTexts** — medical courses and textbooks
- **Surviving Residency** — quick guides for medicine
- **Military Medicine** — doctrinal and instructional field manuals
- **Medical Library** — books on field and emergency medicine
- **NHS Medicines A to Z** — how medicines work and how to take them
- **CDC Travelers' Health** — disease prevention advice
- **Libre Pathology** — a pathology knowledge base
- **Medical Sciences Q&A** — questions and answers on medicine and health
- **WikiVet** — a free veterinary medicine encyclopedia

Knots

Wikipedia's illustrated knot reference, covering essential knots for sailing, climbing, shelter-building, first aid, and general utility.

Wikipedia (Summaries)

The opening summary of every English Wikipedia article, for broad reference across history, science, geography, technology, survival topics, and general knowledge. It holds each article's introduction, not the full text. Useful when you need a wide baseline of information without internet access.

Water Treatment Library

A technical reference covering water sourcing, purification methods (boiling, chemical treatment, filtration, UV), and field water quality assessment, in a collection of PDF guides.

Learning Self Reliance

A practical guide to self-reliance skills including food storage, homesteading, gardening, animal husbandry, and long-term preparedness planning. Focused on building sustainable systems that function without grid dependency.

Survivor Library

An extensive collection of pre-industrial skills, trades, and technical knowledge — covering farming, blacksmithing, construction, medicine, engineering, and more. Designed for long-term resilience when modern supply chains and infrastructure are no longer available.

Food for Preppers

A practical reference for long-term food storage, preservation, and emergency nutrition planning — covering shelf-stable food selection, rotation schedules, caloric planning, and preservation methods including canning, dehydrating, and vacuum sealing.

Project Gutenberg — General Works (9GB)

Over 2,900 public domain volumes, mostly classic magazines and journals such as Punch, Notes and Queries, Harper's, and The Atlantic Monthly, plus volumes of the Encyclopaedia Britannica. Load it in Kiwix to browse and read fully offline. Useful for extended shelter-in-place scenarios, historical reference, and morale during long emergencies.

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SEAMLESS

Available on your Survival Node.

What it is: Seamless is an on-device voice translation app powered by Meta's Unity model from Hugging Face — no internet required. It performs real-time spoken language translation directly on the device, with no data sent to any server. Useful for communicating across language barriers during emergencies or disaster response when no connectivity is available.

Key features:

- Fully offline — all translation runs on-device using the Unity model
- Voice input — speak in one language, hear the translation in another
- No account, no cloud, no data collection
- Supports a wide range of languages

Basic usage:

1. Open **Seamless** from the home screen
2. Select the **source language** (what you're speaking)
3. Select the **target language** (what you want output in)
4. Tap the microphone and speak — translation plays back automatically

Tip: test your target languages before an emergency to confirm the model performs well for your use case.

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DEEPL

Available on your Survival Node.

What it is: DeepL is a high-accuracy text translation app supporting 30+ languages. It is well regarded for producing more natural, context-aware translations than most alternatives — particularly useful for technical, medical, or nuanced content.

Best uses on Survival Node:

- Translating written text — documents, signs, written communications
- Translating longer passages where accuracy and natural phrasing matter
- Complementing Seamless (voice) with high-quality written output

Basic usage:

1. Open **DeepL** from the home screen
2. Type or paste text into the input field
3. Select source and target languages
4. Translation appears immediately below

Note: DeepL needs an internet connection. Use Seamless for fully offline voice translation.

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MORSE

Available on your Survival Node.

What it is: Morse is a simple, fully offline Morse code encoder and decoder. It converts text to Morse code and Morse code back to text — useful for off-grid signaling, radio communication, or emergency signaling with a flashlight or audio tone.

Best uses on Survival Node:

- Encoding messages into Morse for radio or light-based signaling
- Decoding received Morse signals back into readable text
- Learning or refreshing Morse code before an emergency
- SOS signaling reference — three dots, three dashes, three dots (· · · — — — · · ·)

Basic usage:

1. Open **Morse** from the home screen
2. Type your message to encode it into Morse code
3. Or enter Morse dots and dashes to decode back to text
4. Use the audio playback or flashlight output to transmit if supported

Tip: memorize SOS (· · · — — — · · ·) — it is the universal distress signal recognized across all emergency response systems.

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EXODUS PRIVACY

Available on your Survival Node.

What it is: Exodus Privacy is an app auditing tool that scans every app on your Survival Node and reveals which trackers, analytics libraries, and permissions each one uses. It helps you maintain the integrity of your privacy setup over time — especially after installing new apps.

Best uses on Survival Node:

- Audit any new app you install via F-Droid or APK before trusting it
- Verify the privacy posture of your existing app stack
- Identify and remove apps with excessive permissions before a mission

Basic usage:

1. Open **Exodus Privacy** from the home screen
2. The app automatically scans all installed applications
3. Tap any app to see its full report: trackers found, permissions requested, and severity ratings
4. Use this information to decide whether to keep, restrict, or uninstall a given app

Tip: run Exodus after installing any new app to confirm it is clean before using it with sensitive information.

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BRIAR MESSENGER

Available on your Survival Node.

What it is: Briar is an end-to-end encrypted messaging app designed for hostile network environments. It communicates device-to-device via Bluetooth or Wi-Fi — no internet, no central server, no phone number required. When cell networks and internet are down, Briar keeps working.

Key features:

- **Bluetooth messaging** — direct device-to-device with no network at all
- **Wi-Fi messaging** — works across a local network or hotspot with no internet
- End-to-end encrypted — messages are unreadable to any third party
- No phone number or account registration required
- Messages stored on-device only — no cloud backup

Basic usage:

1. Open **Briar** and create a local account (username + password, stored on-device only)
2. To add a contact: tap **+** → **Add contact nearby** — both devices must be open and physically close to scan each other's QR code
3. Enable **Bluetooth** and/or connect both devices to the same Wi-Fi network
4. Tap a contact name to open a chat and send messages
5. Messages deliver when the recipient device comes within Bluetooth or Wi-Fi range

Add your key contacts to Briar before an emergency — contact exchange requires both devices to be present at the same time to scan QR codes.

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MESHTASTIC

Available on your Survival Node. Radio not included.

What it is: Meshtastic is an open-source app for LoRa mesh radios. Paired with a Meshtastic radio over Bluetooth, it sends text messages and GPS locations between radios with no cell service or internet. Messages can hop from radio to radio to extend their range.

Radio required, not included. Only the app is preinstalled. It can't send or receive anything on its own: you need a separate Meshtastic-compatible LoRa radio, and the people you message need radios too.

Basic usage:

1. Turn on your Meshtastic radio, and turn on **Bluetooth** on the phone
2. Open **Meshtastic** and connect to your radio over Bluetooth
3. Set the radio's **region** to the United States. It can't communicate over the mesh until the region is set
4. Send a message to other radios in range

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LOCALSEND

Available on your Survival Node.

What it is: LocalSend is an open-source file transfer app that moves files between nearby devices over a local Wi-Fi network — no internet, no cloud, no USB cable. Works across Android, iOS, Windows, macOS, and Linux.

Best uses on Survival Node:

- Transfer maps, documents, or reference files to another device in the field
- Share medical info, contacts, or plans with others in your group
- Move files between your Survival Node and a laptop over a local hotspot

Key features:

- No internet required — works entirely over local Wi-Fi or device hotspot
- End-to-end encrypted transfers
- Transfers files, folders, text, and clipboard content
- Full Wi-Fi transfer speed — no server throttling
- No login or account needed

Basic usage:

1. Open **LocalSend** on both devices
2. Connect both devices to the same Wi-Fi network (or create a hotspot on one and connect the other to it)
3. LocalSend auto-discovers other nearby devices running the app
4. On the sending device: tap **Send**, select your files, then tap the recipient device name
5. On the receiving device: accept the transfer — files save to your Downloads folder

Tip: if devices don't see each other, confirm they're on the same Wi-Fi network and that LocalSend has local network permission enabled in Android settings.

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SOS FLASHLIGHT

Available on your Survival Node.

What it is: SOS Flashlight turns your phone's built-in flashlight into a bright, automatic SOS distress beacon — flashing the international SOS signal (· · · — — — · · ·) without any manual timing. No internet required, no setup needed.

Best uses on Survival Node:

- Signaling rescuers or search teams at night or in low-visibility conditions
- Attracting attention from aircraft, boats, or distant responders
- Use when voice or radio contact isn't possible

Basic usage:

1. Open **SOS Flashlight** from the home screen
2. Tap the SOS button — the flashlight begins pulsing the SOS pattern automatically
3. Point the phone toward the sky or in the direction of potential rescuers
4. Tap again to stop

Tip: keep your screen brightness low while running SOS Flashlight to conserve battery — the flashlight draw is the primary power use.

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FOSSIFY FILE MANAGER

Available on your Survival Node.

What it is: Fossify File Manager is a lightweight, open-source file manager with no ads, no trackers, and no internet permissions. It gives you full access to your device's local storage — browse, move, copy, rename, and delete files without any cloud involvement.

Best uses on Survival Node:

- Navigating to pre-loaded documents and PDFs (including the Users Guide standard PDF backup in Downloads)
- Moving files received via LocalSend to organized folders
- Managing downloaded AI model files for PocketPal
- Accessing and organizing photos, exports, and offline reference materials

Basic usage:

1. Open **Fossify File Manager** from the home screen
2. Browse your internal storage folders using the left sidebar or main view
3. Long-press any file to select it, then use the toolbar to move, copy, rename, or delete
4. Use the search icon to find files by name across all storage

Tip: the Users Guide standard PDF and the PocketPal system prompt are pre-loaded in your Download folder — open Fossify File Manager to find and read them offline at any time. The Users Guide is also available as an app on the home screen.

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IFIXIT

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*Available on your Survival Node via **Kiwix**.*

What it is: iFixit is a comprehensive repair guide library covering thousands of mechanical and electrical devices — phones, laptops, appliances, tools, and more. It is loaded as a Kiwix library (ZIM file) and is fully available offline, with step-by-step teardown instructions, photos, and parts lists.

Best uses on Survival Node:

- Diagnosing and repairing your own devices when parts or service aren't available
- Field repairs on electronics, generators, radios, and appliances during grid-down scenarios
- Understanding component-level failures before sourcing replacement parts
- Reference for anyone doing their own maintenance without access to a technician

Basic usage:

1. Open **Kiwix** from the home screen
2. Select the **iFixit** library from your downloaded content
3. Search by device name, model, or repair type
4. Select a guide and follow the numbered steps with photos
5. Use the parts and tools list at the top of each guide to prepare before starting

Tip: browse guides for your most critical devices before an emergency so you know what tools and spare parts are worth keeping on hand.

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ACCUBATTERY

Available on your Survival Node.

What it is: AccuBattery is a battery health and diagnostics app. Eastern Shore AI uses it for the one battery health test each unit gets before shipping, and the result is linked in your shipping email.

AccuBattery remains installed on your device and is available if you want to view live charge stats, check battery health, or monitor charge speed — but no testing is required from you.

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EMP PROTOCOL

What is a Faraday Bag?

Manufacturer-rated (Mission Darkness, independent lab tested to MIL-STD-188-125-2 and IEEE 299-2006): 80+ dB attenuation, low MHz to 40 GHz. Source

A Faraday bag is a shielded enclosure lined with conductive material — your Survival Node uses **TitanRF Faraday fabric** — that blocks electromagnetic signals from passing through. Inside a properly sealed Faraday bag, your electronics are shielded from:

- Electromagnetic Pulse (EMP) events and nuclear burst EMP
- Solar flares and geomagnetic storms (Coronal Mass Ejections)
- RF tracking, cellular location signals, Wi-Fi, and Bluetooth
- RFID skimming

A Faraday bag does *not* protect against physical damage, water immersion, or static discharge from improper handling.

EMP protection is only as good as your sealing discipline.

- At first sign of risk, isolate all mission-critical electronics in Faraday bags immediately.
- Fold the opening over itself at least twice before sealing the Velcro.
- Check for punctures, worn seams, or contaminated seal surfaces.
- Keep spare storage sleeves in the case as backup protection.

Note: Every Survival Node includes two Mission Darkness Faraday bags — one for the phone, one for the solar battery.

TROUBLESHOOTING

Before deep debugging, run quick checks in this order:

- Battery level and charging cable integrity
- Storage capacity and free space
- App permissions and recent setting changes
- Full reboot, then retry core apps

If behavior persists, document what changed right before failure. That usually reveals root cause faster than random toggling.

AI giving wrong or unexpected answers? This is normal — models can "hallucinate" (state inaccurate information confidently). Start a fresh chat session to clear context, try rephrasing your question, and always verify critical information against offline reference materials. See the PocketPal section for more on AI limitations.

Battery or device feels unusually hot? Stop charging immediately and move the device to a cool, shaded location. Do not resume charging until it returns to room temperature. If the phone back is warped or deformed, see the Power & Solar Charging section for swollen battery guidance.

App not working or crashing? Some apps that require Google Play Services may not function correctly on debloated Android OS — this is expected behavior. Your core offline preparedness apps (maps, AI, reference tools) are not affected. Contact support@easternshore.ai if a core app fails to function.

Phone won't power on? Connect to the Solar Power Hub and charge for at least 15 minutes before attempting to power on. A deeply discharged lithium battery may need several minutes of charging before the device will respond.

WARRANTY, LEGAL & SUPPORT

Eastern Shore AI, LLC Warranty: 30-day limited warranty on products included in this kit, starting from the delivery date.

Original Manufacturer Warranty: Every Survival Node is built on a hand-vetted smartphone selected for battery life, memory performance, and overall suitability before shipment. The original manufacturer's warranty, if any, is non-transferable and may be void due to OS modification. Eastern Shore AI, LLC's 30-day warranty is your active coverage.

Returns & Refunds (30-Day Window): You may return your Survival Node within 30 days of the delivery date for a refund. To initiate a return, contact us at support@easternshore.ai to receive an RMA (Return Merchandise Authorization) number before shipping. Items returned without an RMA or after 30 days may not be accepted. Once your RMA is issued, ship the item to:

Eastern Shore AI, LLC
PO BOX 317
Parsonsburg, MD 21849

Return shipping costs are the buyer's responsibility. There is no restocking fee. Refunds are issued to the original payment method within 10 business days after the return is received and inspected. Exchanges are not offered.

Right to Refuse & Order Cancellation: Eastern Shore AI, LLC reserves the right to refuse or cancel any order at its discretion, including cases of suspected fraud, misrepresentation, or policy violations. If your order is canceled, you will be notified and fully refunded.

Software Notice: Every Survival Node ships with a deep-debloated Android OS, optimized for performance and longevity. Google Play Services, carrier apps, and known tracking and battery-draining software are removed or disabled. Pre-installed software is provided "AS IS." Some apps that depend on Google Play Services may not function as expected — this does not affect the offline preparedness stack. Motorola may install Android updates when the phone is on Wi-Fi. Only connect to Wi-Fi you trust.

Meshtastic Radio Not Included: The Meshtastic app is preinstalled on the phone, but no Meshtastic or LoRa radio is included with any Survival Node. The app cannot send or receive messages without a separate compatible radio, which must be purchased separately. Eastern Shore AI, LLC does not sell, support, or warrant radio hardware and makes no representation about the compatibility, range, or legal use of any radio you choose to use.

Product Appearance: Hard storage cases are sourced from multiple suppliers — color, latch style, and interior layout may vary slightly from the case shown in product photos. All units are tested and configured to the same functional specifications before shipment.

Shipping: Due to federal regulations governing lithium-ion batteries and cellular devices, all orders ship via USPS ground transportation only — air shipping is not available. Order processing takes up to 7 business days (average 1–2 days). Shipping then takes up to 10 additional business days (average 5–7 days) because of the ground-only requirement. Orders ship to continental U.S. destinations only.

Payment Processing: All transactions are processed securely through Stripe. Eastern Shore AI, LLC does not store payment card information.

Liability: To the maximum extent permitted by law, Eastern Shore AI, LLC is not liable for incidental or consequential damages from product use or inability to use.

Trademark/Affiliation: Eastern Shore AI, LLC is an independent seller and is not an authorized dealer or affiliate of OEM brands referenced for identification purposes only.

Support: Reach us any time at support@easternshore.ai.